

Spark61850

Spark61850 is an IEC 61850 MMS client driver for substation IEDs.

Version 0.1.1 Ignition 8.1.33+ and 8.3 Runs on: any gateway that reaches the substation network

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1. Overview

Spark61850 is a standalone device driver, licensed and running independently of any other SparkBridge module.
Module id `com.greenguardusa.spark61850`.

2. What You Need

- Ignition 8.1.33 or newer, or the 8.3 line (separate build).
- An IED reachable over MMS on ISO (RFC1006 TPKT/COTP), TCP port 102.

3. Install

Install the signed module through Config > Modules > Install or Upgrade a Module; no build step is required.
Module id `com.greenguardusa.spark61850`.

4. License

Licensed per gateway, bought once and kept, soft-licensed, entitlement `com.greenguardusa.spark61850`.

From release 2.7.0, every SparkBridge module runs on Ignition's standard two-hour module trial (the gateway's Reset Trial button restarts the clock). A verified key overrides the trial entirely and the trial clock is then ignored. Drop the key at `<data>/sparkbridge-license.key`; each module re-checks the file every 30 seconds, so installing, replacing, or removing a key takes effect with no restart.

5. Configure

Config > OPC UA > Device Connections > Create new Device > the 61850 device type. Addressing follows the IED/LD/LN/DO/DA model tree, browsable as folders, with functional constraint (ST/MX/CO/SP) as a leaf suffix. SCL file (.icd/.cid/.scl) import can pre-build the model; see the README in the bundle for the exact import steps.

6. Operate

Buffered report reception (RCB) delivers changes as they are published by the IED rather than by polling; watch report-control-block status if updates stop arriving. Controls (SBO and direct) are documented in the README in the bundle.

7. Verify

The MMS/ISO wire encoding has been verified against a real libiec61850 server for Initiate, GetNameList, GetVariableAccessAttributes, Read, Write, and report-control-block (RCB) report reception. A configured Ignition device against a libiec61850 server, browsing and reading through OPC UA including buffered report reception, has passed live on both the 8.1 and 8.3 lines.

8. Troubleshoot

Symptom	Cause / fix
Model tree does not match the IED	Import the IED's SCL file (.icd/.cid/.scl) to pre-build the model rather than relying on live browse alone; see the README for import steps.
Control operation behaves unexpectedly	Confirm whether the control is SBO or direct per the README before issuing a write.

9. Reference

Module id	com.greenguardusa.spark61850
Default port	TCP 102 (MMS/ISO)
SCL import formats	.icd, .cid, .scl
License entitlement	com.greenguardusa.spark61850

10. Support

Docs page section: </sparkbridge/docs#spark61850>. Contact: </sparkbridge/contact>. Release downloads and checksums: [GitHub releases](#).